

UI/UX Design Document

Course Information

Course Title: Software Development III

Course Code : CSE4104

Submitted By

Name :	ID :
<i>Md. Jubayer Hossain</i>	<i>11230121152</i>
<i>Zemima Khan Jimu</i>	<i>11230121144</i>
<i>Orin Khan Joty</i>	<i>11230121147</i>
<i>Rafiuzzaman Rafi</i>	<i>11230121154</i>

Submitted To

Name : Md. Riaz Mahmud

Designation :

Assistant Professor

Department of Computer Science & Engineering

Northern University Bangladesh

Submission Date: 22/06/26

Team Information

Field	Information
Team Name	CSE4104-7C-T03
Section	7C
Project Title	AI Study Material Generator
Team Leader Name	Md. Jubayer Hossain
Team Leader ID	11230121152
Member 1	Zemima Khan jimu ID-11230121144
Member 2	Orin Khan Joty ID -11230121147
Member 3	Rafiuzzaman Rafi ID-11230121154
GitHub Repository Link	https://github.com/jubayerhossain5618/AI-Study-Material-Generator

1. Project Information

Project Title

AI Study Material Generator

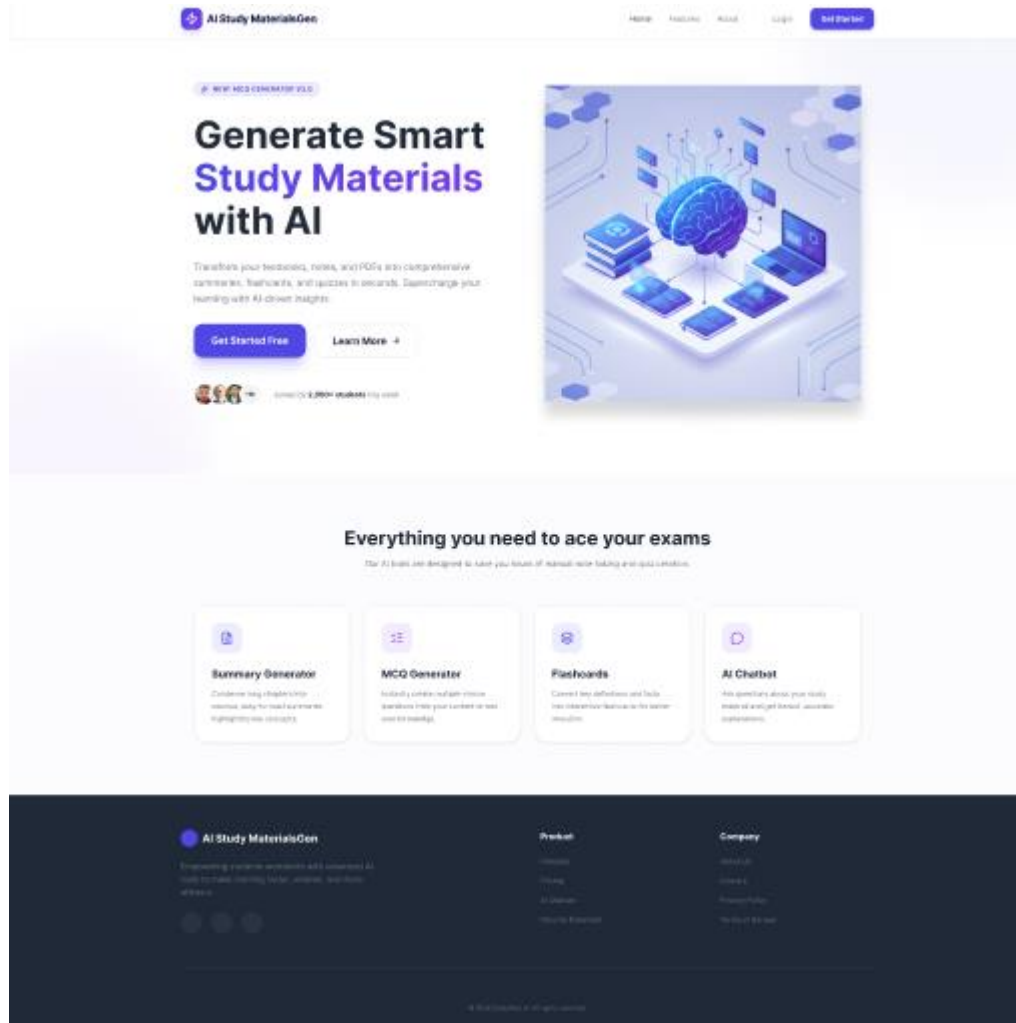
Project Description

The AI Study Material Generator is an intelligent educational platform that automatically converts uploaded study materials into useful learning resources such as summaries, MCQs, flashcards, quizzes, and AI-powered chatbot responses.

The system aims to reduce manual study preparation time and improve learning efficiency through AI integration.

2. UI / Wireframe Designs

Screen 1: Landing / Home Page



Screen 2: Login Page

 **AI Study MaterialsGen**

Welcome Back

Please enter your details to sign in

Email Address

Password [Forgot Password?](#)

☐ Remember me for 30 days

Sign In

OR CONTINUE WITH

Don't have an account? [Sign up for free](#)

Screen 3: Registration Page



Create Account

Join 20,000+ students learning smarter

Full Name

Email Address

Password

Confirm

☐ I agree to the [Terms of Service](#) and [Privacy Policy](#)

Create Account

OR SIGN UP WITH

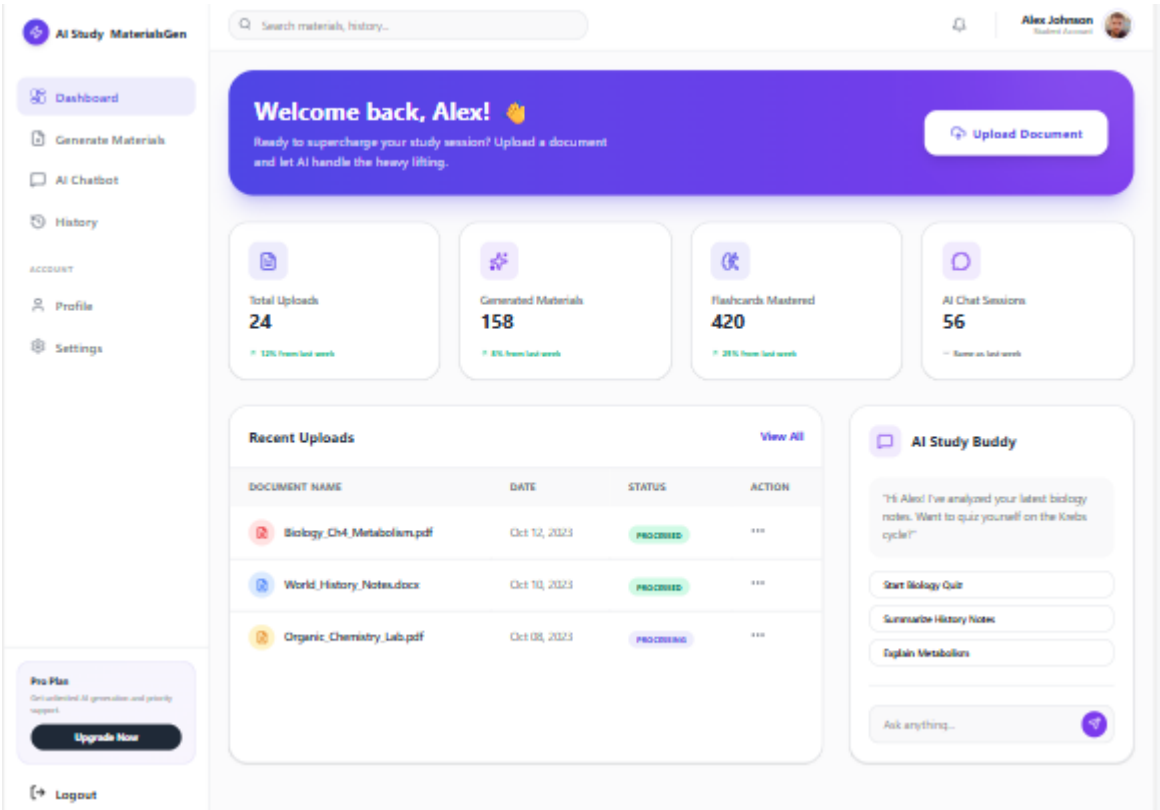
Already have an account? [Sign in](#)



Start Your AI Journey Today

Create an account and get access to powerful study tools that adapt to your learning style.

Screen 4: User Dashboard



Screen 6: AI Material Generation Page

AI Study MaterialsGen

Dashboard

Generate Materials

AI Chatbot

History

ACCOUNT

Profile

AI Material Generator

Credits: 10 remaining

Full

Upload your study material

Support for PDF, DOCX, and TXT (Max 50MB)

Drag and drop your file here or [browse files](#)

BIOCHEMISTRY_LECTURE_NOTES.PDF75%

SELECT OUTPUT TYPE

Summary

MCQs

Flashcards

Full Quiz

Generate Study Materials

Generated SummaryMCQs (12)Flashcards (24)

Biochemistry Summary

Download PDF

Introduction to Metabolism: Metabolism is the set of life-sustaining chemical reactions in organisms. The three main purposes of metabolism are: the conversion of food to energy to run cellular processes; the conversion of food/fuel to building blocks for proteins, lipids, nucleic acids, and some carbohydrates; and the elimination of metabolic wastes.

Anabolism vs Catabolism: Metabolic reactions may be categorized as catabolic – the breaking down of compounds (for example, the breaking down of glucose to pyruvate by cellular respiration); or anabolic – the building up (synthesis) of compounds (such as proteins, carbohydrates, lipids, and nucleic acids).

- **Enzymes:** Biological catalysts that speed up chemical reactions.
- **ATP:** The primary energy carrier in all living organisms.
- **Metabolic Pathways:** Series of chemical reactions occurring within a cell.

Screen 7: AI Chatbot Page

New Chat

RECENT CONVERSATIONS

Metabolism Concepts

World War II Timeline

Python Loops Explained

LAST 7 DAYS

Calculus Derivatives

Organic Chemistry Quiz

Study MaterialsGen AI Assistant

ONLINE & ANALYZING

Export Chat

Hello Alex! I've finished analyzing your notes on **Metabolism**. I noticed you spent some time on the Krebs cycle section.

Would you like me to explain the key inputs and outputs of the cycle, or should we jump straight into a quick 5 question quiz to test your retention?

JUST NOW

Explain the inputs and outputs first, please. A summary table would be great!

SENT

Certainly! Here is a concise breakdown of the Krebs cycle (Citric Acid Cycle) per one turn:

Input	Output
1 Acetyl-CoA	2 CO_2
3 NAD ⁺	3 NADH
1 FAD	1 FADH_2
1 ADP + P _i	1 ATP (or GTP)

Remember, for every **one glucose molecule**, the cycle turns twice!

Generate 5 MCQs

Explain ATP synthesis

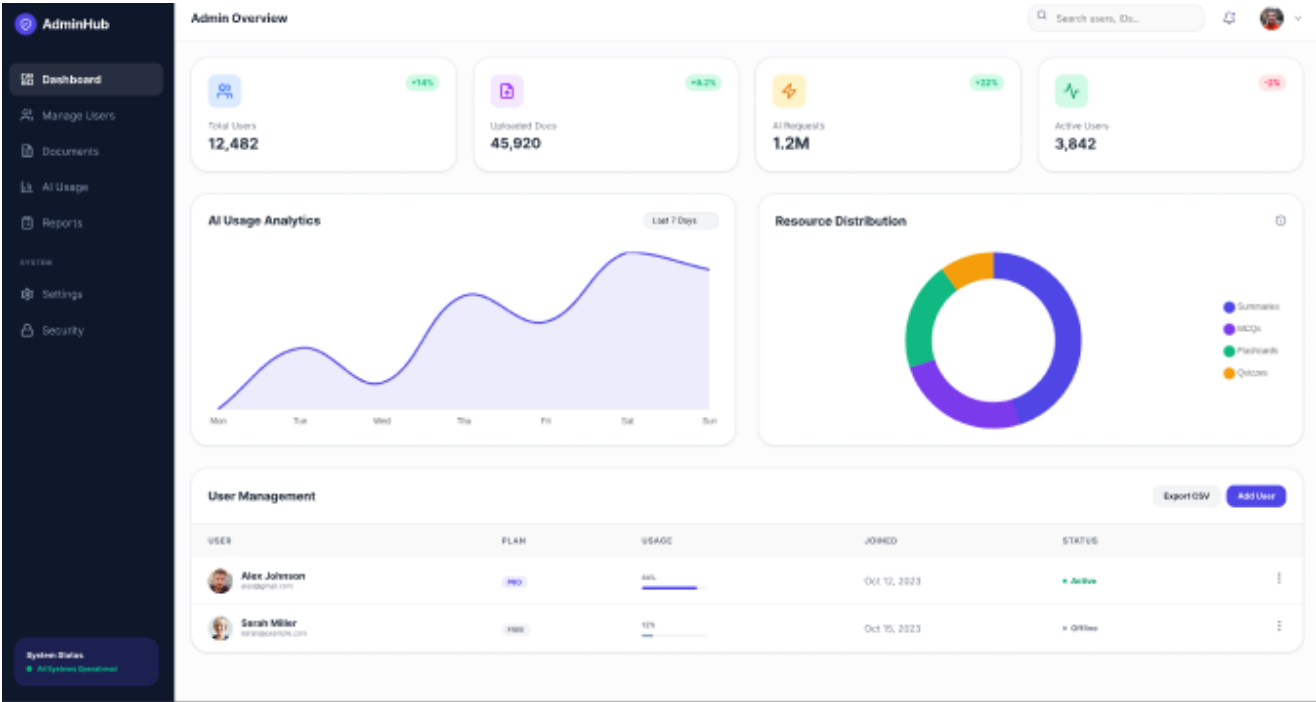
Create Flashcards

Summarize Ch 4

Type your question here...

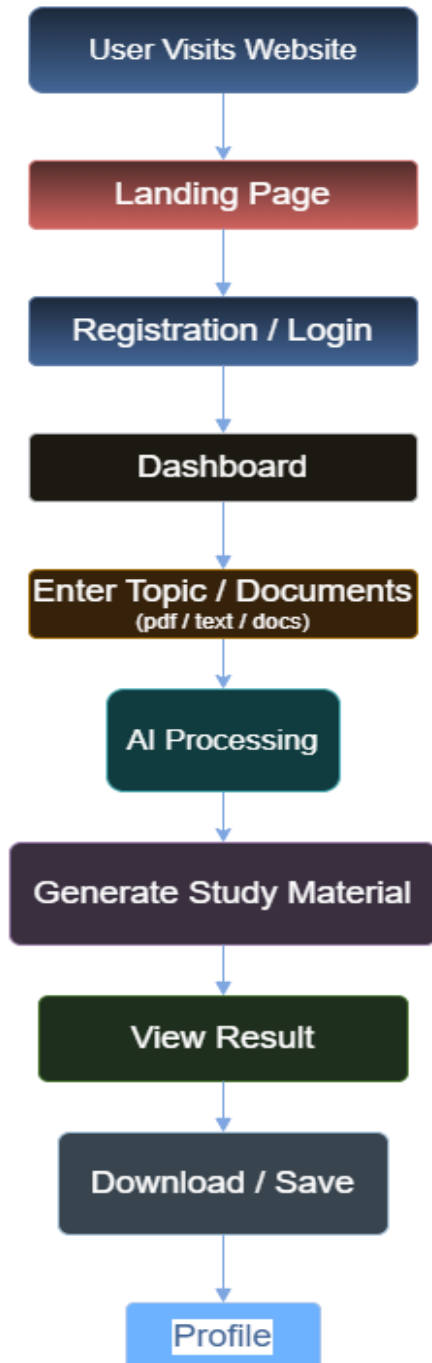
STUDYGEN AI CAN MAKE MISTAKES. VERIFY IMPORTANT INFORMATION.

Screen 8: Admin Dashboard



3. User Flow Diagram

User Flow Diagram AI Study Material Generator



4. Development Roadmap

Week	Development Task
Week 05	Project Setup, UI Design Implementation
Week 06	Backend Development
Week 07	Frontend Development
Week 08	Database Integration
Week 09	AI Integration (OpenAI)
Week 10	Feature Completion
Week 11	Testing and Debugging
Week 12	Deployment Preparation
Week 13	Final Testing and Documentation
Week 14	Final Presentation

5. Team Task Distribution

Name	Role	Responsibility
Md. Jubayer Hossain	Team Leader	Project Management, AI Integration, Testing, Documentation
Zemima Khan jimu	Member 1	Frontend Development (React UI)
Orin Khan Joty	Member 2	Backend Development
Rafiuzzaman Rafi	Member 3	Database Design and API Development

Shared Responsibilities

- GitHub Maintenance
- Testing
- Presentation Preparation

6. Final Technology Stack

Frontend

- HTML
- React.js
- Tailwind CSS
- JavaScript

Backend

- Node.js
- Express.js

Database

- MongoDB

AI Services

- OpenAI API

Authentication

- JWT Authentication

Deployment

Frontend

- Vercel

Backend

- Render

Database

- MongoDB Atlas

Version Control

- Git
 - GitHub
-